
Improving Conditional VAE for Dementia Stage MRI Generation using U-Net Skip Connections and Classifier Guidance

Heesu Choi
2022250043

Wan Arif Fahmi
2024320123

Nur Asha Syafiqah
2023320324

Problem & Proposal

Problem - Vanilla CVAE

Vanilla CVAE can generate or reconstruct brain MRI images, but it mainly focuses on making the image look visually similar.

It has three main problems:

1. **Weak dementia-stage control** - The generated MRI may not clearly match the intended dementia stage.
2. **Loss of important brain details** - Vanilla CVAE may produce blurry images and lose fine anatomical structures.
3. **Less medically meaningful output** - The image may look realistic, but it may not contain strong disease-related features.

Proposal - Classifier-Guided U-Net CVAE

We propose using a classifier-guided U-Net CVAE to improve dementia-stage image generation.

The model uses:

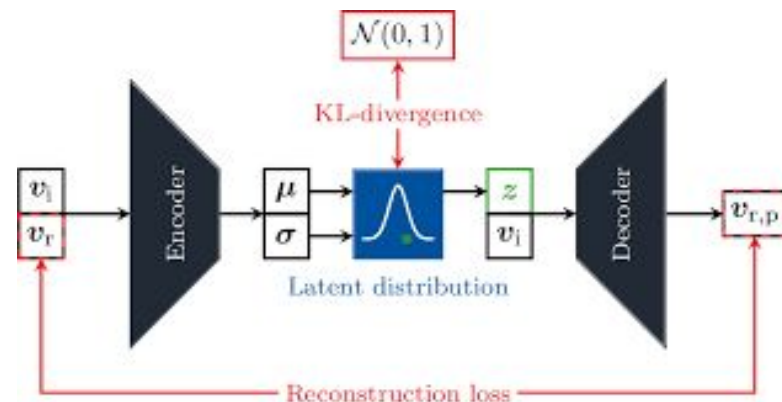
1. **U-Net structure** - Preserve brain anatomy and spatial details through skip connections.
2. **Conditional VAE** - Uses dementia-stages labels to guide image reconstruction.
3. **Frozen classifier guidance** - Checks whether the generated MRI matches the correct dementia stage.

Baseline Model

Baseline Model : Vanilla CVAE

What is the Vanilla CVAE?

- **Definition:** The Vanilla CVAE (Conditional Variational Autoencoder) is a standard conditional generative model.
- **Core Mechanism:** It maps an input MRI slice into a latent representation and reconstructs it based on a given target condition.
- **Objective Function:** It relies strictly on fundamental objectives, specifically Reconstruction Loss and KL Divergence.
- **Key Distinction:** Unlike the proposed model, the Vanilla CVAE completely lacks auxiliary supervision, such as "classification-guided loss" or latent clustering mechanisms, to enforce stage-specific structural changes.



Baseline Model : Vanilla CVAE

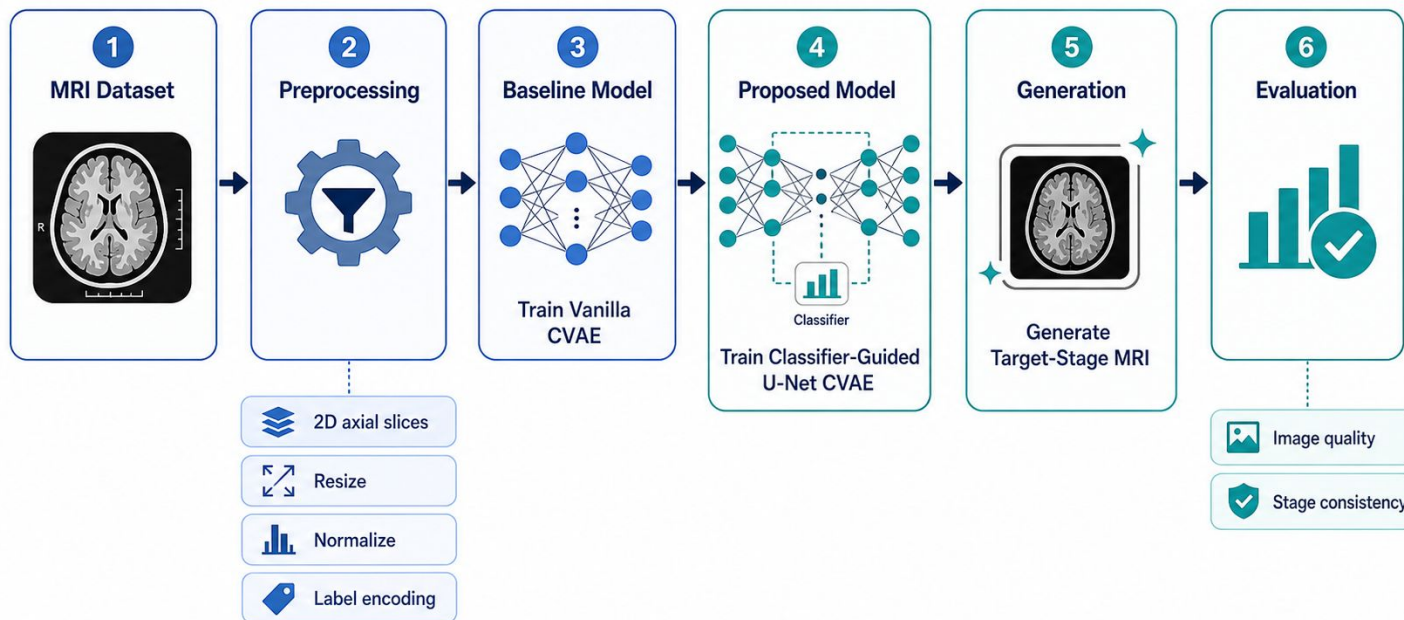
Why Vanilla CVAE?

- **Role in the Experiment:** It serves as the primary control group (baseline) to evaluate the effectiveness of the newly proposed framework.
- **Why it was Adopted:** It was chosen to conduct an ablation study, specifically to isolate and measure the impact of integrating the "Classification-guided Loss" into the generative process.
- **Experimental Purpose:** By comparing the proposed model against this Vanilla baseline, the researchers aimed to prove that standard conditional generation alone is insufficient for precise disease-stage control and semantic consistency in medical imaging.

Methodology

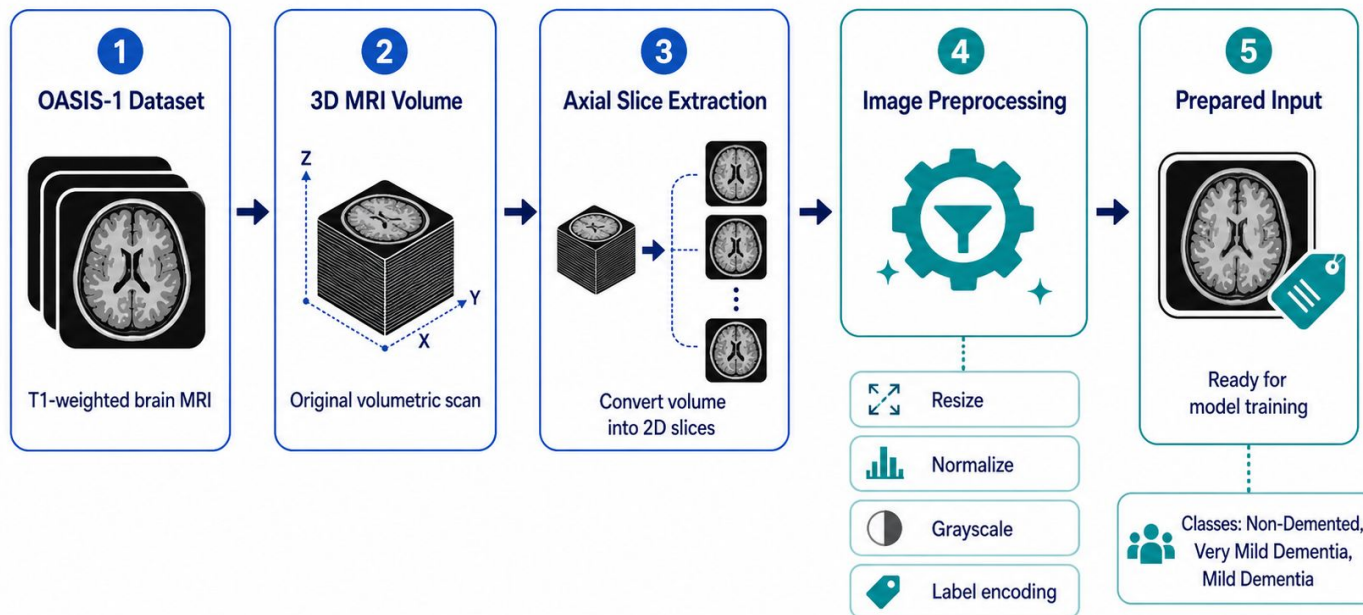
Methodology

Proposed Model: Classifier-Guided U-Net CVAE



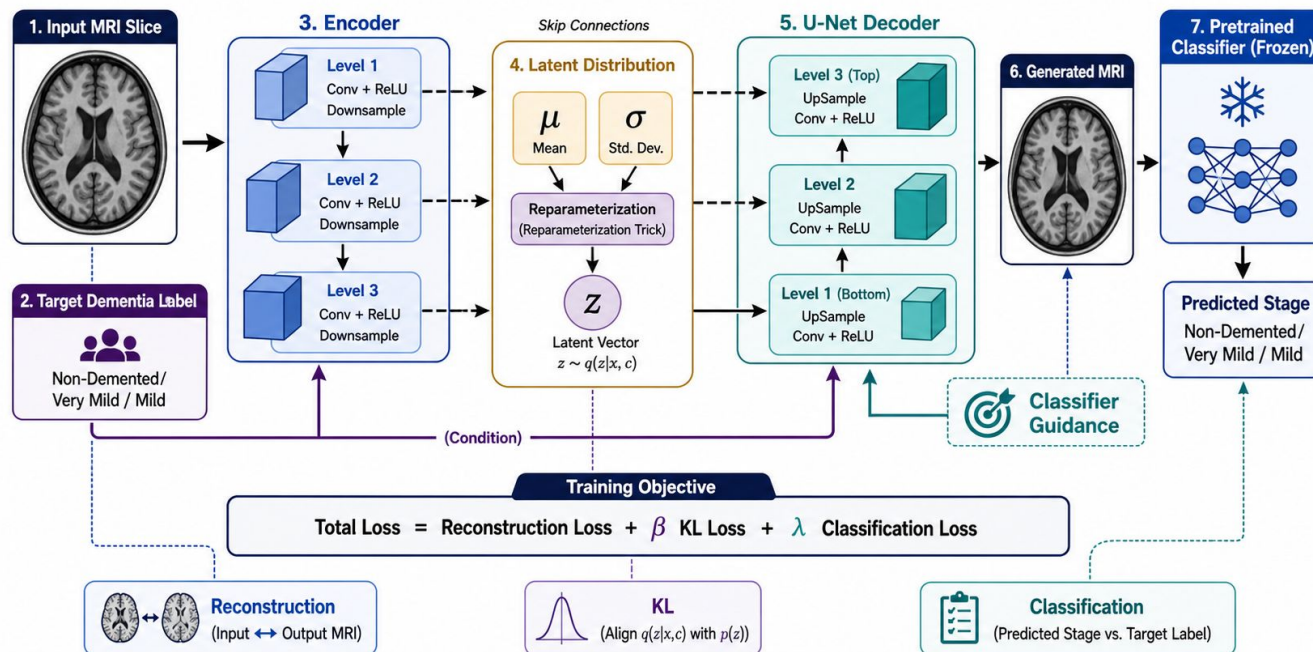
Methodology

Dataset & Preprocessing



Methodology

Proposed Model Architecture



Methodology

Loss Function

$$\text{Total Loss} = \text{Reconstruction Loss} + \beta \text{ KL Loss} + \lambda \text{ Classification Loss}$$

Loss	Purpose
Reconstruction Loss	Makes generated MRI visually close to the input MRI
KL Loss	Keeps the latent space smooth and regularized
Classification Loss	Encourages generated MRI to match the target dementia stage

Methodology

Evaluation Strategy

1. Image Quality

- a. MSE / SSIM / PSNR
- b. Visual comparison of reconstructed MRI

2. Stage Consistency

- a. Target-class confidence
- b. Confusion Matrix

3. Qualitative Explainability

- a. Grad-CAM Heatmap on generated Images

Experiment & Results

Model Comparison

Aspect	UNet CVAE	Vanilla CVAE
Input	MRI image	MRI image + label map
Encoder	Conv encoder with skip features	Conv encoder only
Latent space	$z = 128$	$z = 128$
Conditioning	Label embedding at bottleneck	One-hot label at bottleneck
Decoder	Upsampling + conv blocks	Transposed convolution
Skip connections	Yes	No
Output	$1 \times 224 \times 224$ reconstruction	$1 \times 224 \times 224$ reconstruction
Loss	BCE + KL + classifier loss	BCE + KL

Experiment

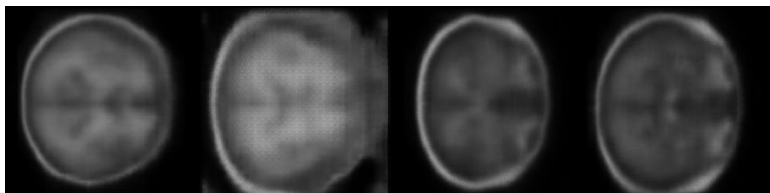
Experimental Setup

Metric	UNet	Vanilla	Interpretation
Images	16,958	16,958	Same validation set size
MSE	0.000103	0.012838	UNet is approximately 124.6× lower
PSNR	39.9667	18.9932	UNet is +20.97 dB higher
SSIM	0.9982	0.7264	UNet is +0.2718 higher
Recon BCE	0.324215	0.362892	UNet is approximately 10.7% lower
KL Loss	0.000004	0.026968	UNet is substantially lower
Val Loss	0.324219	0.389860	UNet is approximately 16.8% lower

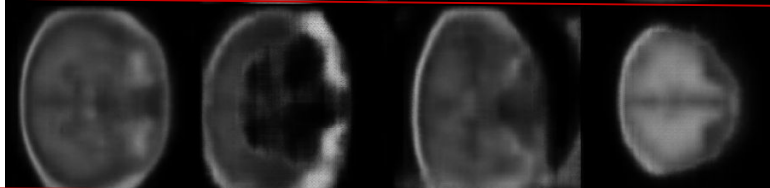
Result

Baseline Result

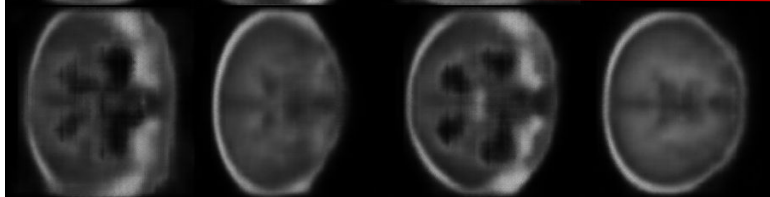
Non-Demented



Very Mild Dementia



Mild Dementia



Confusion matrix:

[13298, 0, 0]

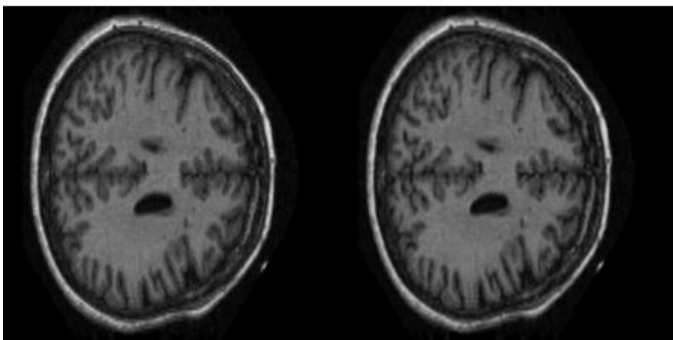
[2806, 0, 0]

[854, 0, 0]

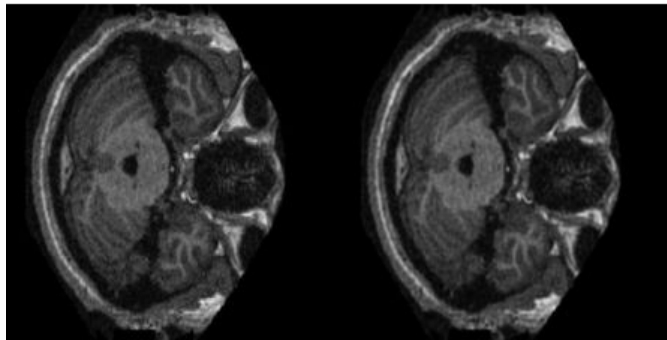
Results

Classifier-Guided U-Net CVAE

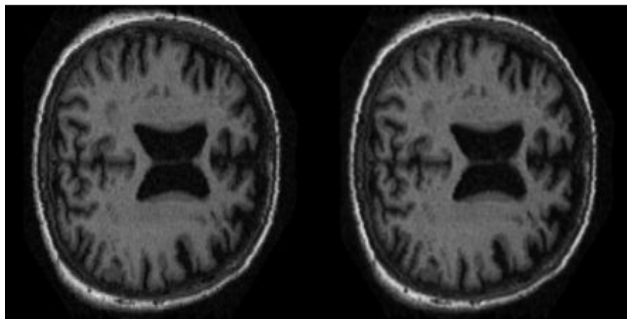
Original | Reconstruction | Very mild Dementia



Original | Reconstruction | Non Demented



Original | Reconstruction | Mild Dementia



Confusion matrix:
[13203, 84, 11]
[375, 2428, 3]
[61, 66, 727]

Future Implementation

- Add target-stage confidence comparison
- Add Grad-CAM visualization
- Implementing FiLM

References

1. Sohn, K., Lee, H., & Yan, X. (2015). *Learning structured output representation using deep conditional generative models*. Advances in Neural Information Processing Systems.
2. Ronneberger, O., Fischer, P., & Brox, T. (2015). *U-Net: Convolutional Networks for Biomedical Image Segmentation*. Medical Image Computing and Computer-Assisted Intervention (MICCAI), 234–241.
3. Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*. IEEE International Conference on Computer Vision (ICCV).
4. Wang, Z., Bovik, A. C., Sheikh, H. R., & Simoncelli, E. P. (2004). *Image quality assessment: From error visibility to structural similarity*. IEEE Transactions on Image Processing, 13(4), 600–612.
5. Kingma, D. P., & Welling, M. (2014). *Auto-Encoding Variational Bayes*. International Conference on Learning Representations (ICLR).

GitHub Link:

- <https://github.com/asha1603/20261R0136COSE47400>

THANK YOU!

